Panel Data, Causal Inference and Difference-in-Differences

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Econometrics II

Lecture Plan

Part I: Introduction to Panel Data

Part I

Introduction to Panel Data

Types of Data Structures

- **Cross-sectional data**: many units ($i = 1, \ldots, n$) observed at a single point in time.

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

- **Time series data**: a single unit observed over many periods ($t = 1, \ldots, T$). This was the focus of most of this course.

$$y_t = \beta_0 + \beta_1 x_t + u_t$$

- **Pooled cross sections**: independent cross-sectional samples drawn at different points in time (different units each period).
- **Panel (longitudinal) data**: the *same* units i are observed over *multiple* periods t.

$$y_{it} = \beta_0 + \beta_1 x_{it} + u_{it}, \qquad i = 1, \ldots, n; \;\; t = 1, \ldots, T$$

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What Does Panel Data Look Like?

- Each observation is identified by two indices: a cross-sectional unit i and a time period t.
- Example: data on n countries (or firms, individuals, cities) followed over T years.

unit (i)	year (t)	y_{it}	x_{it}
1	2018	$y_{1,2018}$	$x_{1,2018}$
1	2019	$y_{1,2019}$	$x_{1,2019}$
1	2020	$y_{1,2020}$	$x_{1,2020}$
2	2018	$y_{2,2018}$	$x_{2,2018}$
2	2019	$y_{2,2019}$	$x_{2,2019}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$

- A **balanced** panel has all T periods for every unit; otherwise it is **unbalanced**.

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Why Use Panel Data?

- The key advantage: panel data lets us control for **unobserved, time-invariant** characteristics of the units.
- Many important variables are hard or impossible to measure: ability, motivation, managerial quality, geography, culture, institutions, "business climate", etc.
- If these omitted factors are correlated with our regressors, OLS on a single cross section is biased (omitted variable bias).
- Idea: if such a factor does not change over time for a given unit, we can **eliminate** it by exploiting the repeated observations on the same unit.
- This is the panel-data route to a more credible *causal* interpretation.

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The Unobserved Effects Model

- Consider the model

$$y_{it} = \beta_1 x_{it} + a_i + u_{it}, \qquad t = 1, \ldots, T$$

  (we suppress the intercept / time dummies for now).
- a_i is the **unobserved effect** (also called the *fixed effect*, *individual heterogeneity*, or *unobserved heterogeneity*). It is constant over time but differs across units.
- u_{it} is the **idiosyncratic error**: it varies across both i and t.
- The problem: a_i is part of the error and may be correlated with x_{it},

$$\text{Cov}(x_{it}, a_i) \neq 0,$$

  so pooled OLS (treating $a_i + u_{it}$ as one composite error) is biased and inconsistent.

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Two Periods: First Differencing

- Suppose $T = 2$ (periods $t = 1, 2$). Write the model with a time intercept:

$$y_{it} = \beta_0 + \delta_0 d2_t + \beta_1 x_{it} + a_i + u_{it}$$

  where $d2_t = 1$ in period 2 and 0 in period 1.
- Subtract the period-1 equation from the period-2 equation for the same unit i:

$$\underbrace{(y_{i2} - y_{i1})}_{\Delta y_i} = \delta_0 + \beta_1 \underbrace{(x_{i2} - x_{i1})}_{\Delta x_i} + \underbrace{(u_{i2} - u_{i1})}_{\Delta u_i}$$

- The unobserved effect a_i **drops out**! We can then estimate

$$\Delta y_i = \delta_0 + \beta_1 \Delta x_i + \Delta u_i$$

  by OLS. This is the **first-differenced (FD) estimator**.

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First Differencing: Assumptions

- For the FD estimator to be unbiased / consistent we need the differenced regressor to be uncorrelated with the differenced error,
$$\text{Cov}(\Delta x_i, \Delta u_i) = 0,$$
which follows from the strict exogeneity condition $\text{E}(u_{it} \mid x_{i1}, x_{i2}, a_i) = 0$.
- Crucially, we no longer need a_i to be uncorrelated with x_{it} — a_i has been differenced away. This is the power of panel data.
- We do need Δx_i to have variation across units (some units must change their x). Time-constant regressors cannot be included — they difference out too.
- With $T > 2$ periods, FD is applied to successive differences; the within (fixed effects) estimator is the more common alternative.

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The Fixed Effects (Within) Estimator

- An alternative way to remove a_i with $T \geq 2$ is the **within transformation**.
- Start from $y_{it} = \beta_1 x_{it} + a_i + u_{it}$. Average over time for each unit i:
$$\bar{y}_i = \beta_1 \bar{x}_i + a_i + \bar{u}_i, \qquad \bar{y}_i = \tfrac{1}{T}\sum_{t=1}^{T} y_{it}$$
- Subtract the time-average (the "within" demeaning):
$$(y_{it} - \bar{y}_i) = \beta_1(x_{it} - \bar{x}_i) + (u_{it} - \bar{u}_i)$$
- Again a_i is eliminated. OLS on the demeaned data gives the **fixed effects (FE) estimator** $\hat{\beta}_1^{FE}$.
- Equivalently, FE = OLS with a separate dummy variable for each unit (the *Least Squares Dummy Variable*, LSDV, approach).

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Fixed Effects: Remarks

- When $T = 2$, FE and FD give numerically identical estimates. For $T > 2$ they differ (FE is more efficient if u_{it} is serially uncorrelated; FD is preferred if u_{it} is close to a random walk).
- FE allows arbitrary correlation between a_i and the regressors x_{it} — this is its great strength.
- **Cost**: effects of *time-invariant* variables (e.g. gender, race, distance to coast) cannot be estimated; they are absorbed by a_i.
- In practice we also include **time dummies** to control for aggregate shocks common to all units in a period.
- Standard errors should be **clustered** at the unit level to allow for serial correlation within units.

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Random Effects

- The **random effects (RE)** model keeps a_i in the composite error but assumes it is *uncorrelated* with the regressors:
$$\text{Cov}(x_{it}, a_i) = 0 \quad \text{for all } t.$$
- Under this assumption pooled OLS is consistent but inefficient (the composite error $v_{it} = a_i + u_{it}$ is serially correlated). RE uses a (feasible) GLS transformation that quasi-demeans the data:
$$y_{it} - \theta \bar{y}_i, \qquad 0 \leq \theta \leq 1.$$
- RE is more efficient *if* its key assumption $\text{Cov}(x_{it}, a_i) = 0$ holds, and it can estimate coefficients on time-invariant variables.
- But if a_i is correlated with x_{it}, RE is **biased** — exactly the case FE handles.

Fixed Effects vs. Random Effects

- Central question: is the unobserved effect a_i correlated with the regressors?
  - **Yes** $\Rightarrow$ use Fixed Effects (RE is biased).
  - **No** $\Rightarrow$ Random Effects is consistent and more efficient.
- The **Hausman test** compares the FE and RE estimates:

$$H_0: \text{Cov}(x_{it}, a_i) = 0 \quad \text{(RE appropriate)},$$
$$H_1: \text{Cov}(x_{it}, a_i) \neq 0 \quad \text{(use FE)}.$$

  A large difference between $\hat{\beta}^{FE}$ and $\hat{\beta}^{RE}$ leads to rejecting H_0.
- In applied microeconomics, FE is usually the default for causal questions because it is robust to a leading source of endogeneity.

Panel Data: Summary

- Panel data follow the same units over time.
- By differencing or demeaning, we can remove time-invariant unobserved heterogeneity a_i that would otherwise bias OLS.
- First differencing (FD) and fixed effects (FE) are the two workhorse estimators; they coincide when $T = 2$.
- This idea — using *within-unit* variation over time to control for confounders — is the bridge to modern causal inference and to difference-in-differences.

Part II: Introduction to Causal Inference

Part II

Introduction to Causal Inference

Correlation Is Not Causation

- Most of econometrics estimates conditional expectations / correlations. But economic policy questions are usually **causal**: "what is the effect of X on Y?"
  - Does an extra year of schooling *cause* higher wages?
  - Does a minimum wage increase *cause* job losses?
  - Does a job-training program *cause* higher employment?
- A regression coefficient is causal only under strong assumptions. Threats:
  - **Omitted variables / confounding** (the "opera and longevity" example revisited).
  - **Reverse causality / simultaneity**.
  - **Selection** into treatment.
- Causal inference is a set of *research designs* that make the "other things equal" (ceteris paribus) comparison credible.

A Motivating Puzzle: Do Hospitals Make People Healthier?

- A simple question (from Angrist and Pischke, *Mostly Harmless Econometrics*): does going to the hospital make people healthier?
- Compare the self-reported health of people who went to the hospital with those who did not (health on a 1–5 scale: $1 =$ excellent, $\ldots$, $5 =$ poor, so **higher means worse**):

Group	Sample Size	Mean health status	Std. Error
Hospital	7774	2.79	0.014
No Hospital	90049	2.07	0.003

- The difference is $2.79 - 2.07 = 0.72$, statistically significant. Taken at face value, this says hospitals make people **sicker**!
- Clearly something is wrong with this comparison. What?

The Potential Outcomes Framework

- Framework due to Neyman and Rubin (the *Rubin Causal Model*).
- Let $D_i \in \{0, 1\}$ be a binary **treatment** ($1 =$ treated, $0 =$ control / untreated).
- Each unit i has **two potential outcomes**:

  $y_i(1) =$ outcome if treated, $\quad y_i(0) =$ outcome if not treated.
- The **causal effect** of the treatment for unit i is

$$\tau_i = y_i(1) - y_i(0).$$

- We only ever observe one of the two:

$$y_i = D_i\, y_i(1) + (1 - D_i)\, y_i(0).$$

- Hospital example: for person i, $y_i(1)$ is their health *if* hospitalized and $y_i(0)$ is their health *if not*. We see a sick person's $y_i(1)$, but never the $y_i(0)$ for the *same* person at the *same* time.

The Fundamental Problem of Causal Inference

- For any single unit, one potential outcome is observed and the other is the unobserved **counterfactual**.
- We can never compute $\tau_i = y_i(1) - y_i(0)$ for an individual — this is the **fundamental problem of causal inference** (Holland, 1986).
- So we focus on **average** effects across a population:

$$\text{ATE} = \mathsf{E}[y_i(1) - y_i(0)], \qquad \text{ATT} = \mathsf{E}[y_i(1) - y_i(0) \mid D_i = 1].$$

- ATE $=$ average treatment effect; ATT $=$ average treatment effect on the treated.
- Estimating these requires constructing a credible counterfactual for the treated group.

Selection Bias

- The naive estimator is the difference in observed means between treated and untreated:

$$\mathsf{E}[y_i \mid D_i = 1] - \mathsf{E}[y_i \mid D_i = 0].$$

- Decompose it:

$$\mathsf{E}[y_i \mid D_i = 1] - \mathsf{E}[y_i \mid D_i = 0] = \underbrace{\mathsf{E}[y_i(1) \mid D_i = 1] - \mathsf{E}[y_i(0) \mid D_i = 1]}_{\text{ATT (causal)}} + \underbrace{\mathsf{E}[y_i(0) \mid D_i = 1] - \mathsf{E}[y_i(0) \mid D_i = 0]}_{\text{selection bias}}$$

- **Selection bias**: treated and untreated units differ in their untreated potential outcome $y_i(0)$, i.e. they are not comparable to begin with.
- Example: people who enroll in training may have lower earnings prospects anyway; comparing them to non-enrollees understates the program's effect.

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Back to the Hospital Puzzle

- Now the puzzle is easy to explain. The naive difference mixes two things:

$$\underbrace{0.72}_{\text{observed}} = \underbrace{\mathsf{E}[y_i(1) - y_i(0) \mid D_i = 1]}_{\text{true effect of hospital (ATT)}} + \underbrace{\mathsf{E}[y_i(0) \mid D_i = 1] - \mathsf{E}[y_i(0) \mid D_i = 0]}_{\text{selection bias}}.$$

- **Selection bias here is large and positive**: people who go to the hospital are sicker to begin with, so even untreated they would have worse health, $\mathsf{E}[y_i(0) \mid D_i = 1] > \mathsf{E}[y_i(0) \mid D_i = 0]$.
- The hospital may truly *improve* health (a negative ATT on this scale), but the selection bias is so large it swamps and even reverses the sign of the observed difference.
- **Lesson**: comparing treated vs. untreated is *not* the same as the causal effect, unless the two groups are comparable in their $y_i(0)$.

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Randomized Experiments: The Gold Standard

- If treatment D_i is assigned **randomly**, it is independent of potential outcomes:

$$\{y_i(0), y_i(1)\} \perp D_i.$$

- Then treated and control groups are comparable on average, so $\mathsf{E}[y_i(0) \mid D_i = 1] = \mathsf{E}[y_i(0) \mid D_i = 0]$ and **selection bias is zero**.
- The simple difference in means consistently estimates the ATE:

$$\widehat{\mathsf{ATE}} = \bar{y}_{\text{treated}} - \bar{y}_{\text{control}}.$$

- Randomized controlled trials (RCTs) are the benchmark, but often infeasible, unethical, or expensive in economics.
- This motivates **quasi-experimental** (observational) designs that mimic randomization.

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A Menu of Identification Strategies

- When we cannot randomize, we look for a source of variation in D_i that is "as good as random":
  - **Randomized experiments (RCTs)** — assignment is literally random.
  - **Instrumental variables (IV)** — a variable affecting D but not y directly.
  - **Regression discontinuity (RD)** — treatment switches at a cutoff of a running variable.
  - **Difference-in-differences (DiD)** — compare changes over time between treated and control groups.
- These are often based on **natural experiments**: policy changes, reforms, or events that affect some units but not others.
- We now study DiD in detail — it connects directly to the panel-data ideas from Part I.

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Part III: Difference-in-Differences

Part III

Difference-in-Differences (DiD)

The Basic Idea

- Setting: a **policy / treatment** affects one group (the **treatment group**) but not another (the **control group**), and we observe both groups **before** and **after** the policy.
- Naive comparison 1 — *before vs. after* for the treated group: contaminated by other things that changed over time (aggregate trends).
- Naive comparison 2 — *treated vs. control after* the policy: contaminated by pre-existing differences between the groups (selection).
- **DiD idea**: combine the two. Use the control group's *change* over time to estimate what *would have happened* to the treated group absent the policy (the counterfactual).
- Take the **difference of the two differences**.

The 2×2 DiD Design

- Two groups (Treatment T, Control C) and two periods (Before, After).
- Let $\bar{y}$ denote the group–period average of the outcome.

	Before	After	After − Before
Treatment	$\bar{y}_{T,B}$	$\bar{y}_{T,A}$	$\Delta_T = \bar{y}_{T,A} - \bar{y}_{T,B}$
Control	$\bar{y}_{C,B}$	$\bar{y}_{C,A}$	$\Delta_C = \bar{y}_{C,A} - \bar{y}_{C,B}$
		DiD:	$\hat{\tau} = \Delta_T - \Delta_C$

- The DiD estimator:

$$\hat{\tau}_{DiD} = (\bar{y}_{T,A} - \bar{y}_{T,B}) - (\bar{y}_{C,A} - \bar{y}_{C,B}).$$

- Time-constant differences between groups and common time trends both cancel out.

DiD as a Regression

- Let $T_i = 1$ for the treatment group, $\text{Post}_t = 1$ in the after period. Estimate

$$y_{it} = \beta_0 + \beta_1 T_i + \beta_2 \text{Post}_t + \beta_3 (T_i \times \text{Post}_t) + u_{it}.$$

- Interpreting the coefficients via the group–period means:

	Before	After
Control	β_0	$\beta_0 + \beta_2$
Treatment	$\beta_0 + \beta_1$	$\beta_0 + \beta_1 + \beta_2 + \beta_3$

- The **interaction coefficient** is the DiD estimate:

$$\boxed{\hat{\beta}_3 = (\bar{y}_{T,A} - \bar{y}_{T,B}) - (\bar{y}_{C,A} - \bar{y}_{C,B}) = \hat{\tau}_{DiD}}$$

- The regression form gives standard errors directly and lets us add covariates.

The Parallel Trends Assumption

- DiD identifies the causal effect **only if** the parallel trends assumption holds:

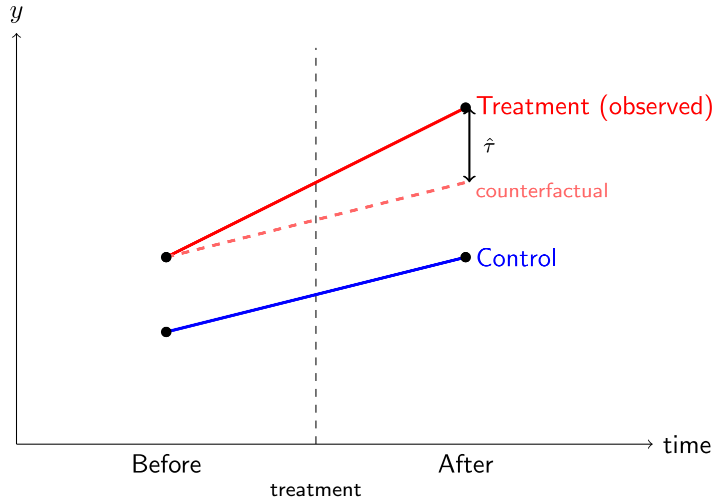
  > *In the absence of treatment, the treatment and control groups would have followed the* ***same trend*** *(same average change) in the outcome over time.*

- Formally, in potential-outcome terms:

$$\mathsf{E}[y_{T,A}(0) - y_{T,B}(0)] = \mathsf{E}[y_{C,A}(0) - y_{C,B}(0)].$$

- The levels of the two groups need *not* be equal — only their counterfactual *trends* must match.
- This assumption is **not directly testable** (the treated counterfactual is unobserved), but it is made more credible by inspecting **pre-treatment trends**.

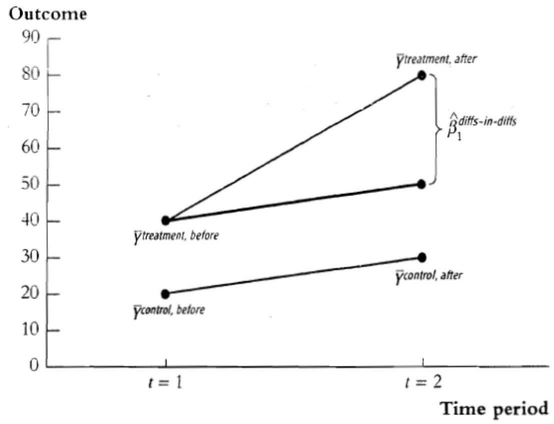
Parallel Trends: Illustration



- The DiD effect $\hat{\tau}$ is the gap between the treated group's *observed* outcome and its *counterfactual* (control trend applied to the treated group).

DiD Estimator in Two Periods: A Numerical Example

The post treatment difference between the treatment and control groups is $80 - 30 = 50$, but this overstates the treatment effect because before the treatment $\overline{Y}$ was higher for the treatment than the control group by $40 - 20 = 20$ The differences-in-differences estimator is the difference between the final and initial gaps, so $\hat{\beta}_1^{\text{diffs-in-diffs}} = (80 - 30) - (40 - 20) = 50 - 20 = 30$. Equivalently, the differences-in-differences estimator is the average change for the treatment group minus the average change for the control group, that is, $\hat{\beta}_1^{\text{diffs-in-diffs}} = \Delta\overline{Y}^{\text{treatment}} - \Delta\overline{Y}^{\text{control}} = (80 - 40) - (30 - 20) = 30$.

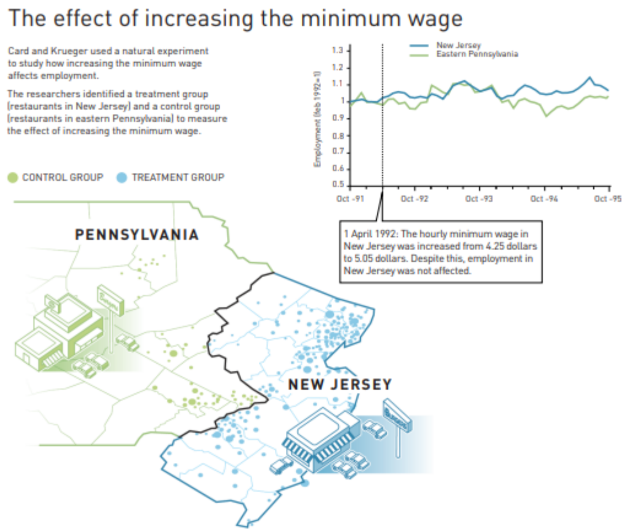


Source: Stock and Watson, *Introduction to Econometrics*, 3rd ed., ch. 13, p. 492.

Classic Application: Minimum Wage and Employment

- Card and Krueger (1994), *Minimum Wages and Employment: A Case Study of the Fast-Food Industry in New Jersey and Pennsylvania*, American Economic Review, 84(4), 772–793.
- On April 1, 1992, New Jersey's minimum wage rose from \$4.25 to \$5.05 per hour. In neighboring Pennsylvania it stayed unchanged.
- Standard competitive theory predicts that a higher wage *reduces* labor demand, so employment should fall.
- They focus on the **fast-food industry**, which employs many low-skilled, minimum-wage workers and is easy to sample comparably across the two states.
- This is a classic **natural experiment**: a policy change affects one group (NJ) but not a comparable neighbor (PA).
- David Card was awarded the 2021 Nobel Memorial Prize in Economics, in part for this work.

The Card and Krueger Natural Experiment



Source: The Nobel Committee for the Prize in Economic Sciences, nobelprize.org (2021).

Card and Krueger: Design and Data

- **Treatment group**: fast-food restaurants in New Jersey (minimum wage rose).
- **Control group**: fast-food restaurants in eastern Pennsylvania (no change).
- Why these states? They are neighbors, exposed to common macroeconomic shocks, and plausibly similar in many unobservable respects — making parallel trends credible.
- They surveyed the same restaurants in **two waves**: Wave I in Feb/March 1992 (*before*) and Wave II in Nov/Dec 1992 (*after*).
- Outcome: **full-time-equivalent (FTE) employment** = number of full-time workers (incl. managers) + $0.5\times$ number of part-time workers.
- DiD = (change in NJ employment) − (change in PA employment).

Card and Krueger (1994): Average Employment

TABLE 3—AVERAGE EMPLOYMENT PER STORE BEFORE AND AFTER THE RISE IN NEW JERSEY MINIMUM WAGE

	Stores by state			Stores in New Jersey[a]			Differences within NJ[b]	
Variable	PA (i)	NJ (ii)	Difference, NJ − PA (iii)	Wage = \$4.25 (iv)	Wage = \$4.26–\$4.99 (v)	Wage ≥ \$5.00 (vi)	Low–high (vii)	Midrange–high (viii)
1. FTE employment before, all available observations	23.33 (1.35)	20.44 (0.51)	−2.89 (1.44)	19.56 (0.77)	20.08 (0.84)	22.25 (1.14)	−2.69 (1.37)	−2.17 (1.41)
2. FTE employment after, all available observations	21.17 (0.94)	21.03 (0.52)	−0.14 (1.07)	20.88 (1.01)	20.96 (0.76)	20.21 (1.03)	0.67 (1.44)	0.75 (1.27)
3. Change in mean FTE employment	−2.16 (1.25)	0.59 (0.54)	2.76 (1.36)	1.32 (0.95)	0.87 (0.84)	−2.04 (1.14)	3.36 (1.48)	2.91 (1.41)
4. Change in mean FTE employment, balanced sample of stores[c]	−2.28 (1.25)	0.47 (0.48)	2.75 (1.34)	1.21 (0.82)	0.71 (0.69)	−2.16 (1.01)	3.36 (1.30)	2.87 (1.22)
5. Change in mean FTE employment, setting FTE at temporarily closed stores to 0[d]	−2.28 (1.25)	0.23 (0.49)	2.51 (1.35)	0.90 (0.87)	0.49 (0.69)	−2.39 (1.02)	3.29 (1.34)	2.88 (1.23)

Notes: Standard errors are shown in parentheses. The sample consists of all stores with available data on employment. FTE (full-time-equivalent) employment counts each part-time worker as half a full-time worker. Employment at six closed stores is set to zero. Employment at four temporarily closed stores is treated as missing.

Source: Card and Krueger (1994), Table 3, p. 780. Standard errors in parentheses.

Card and Krueger: The DiD Estimate

- Average FTE employment per store (Table 3, columns (i)–(ii)):

	PA (control)	NJ (treatment)	NJ − PA
Before (Wave I)	23.33	20.44	-2.89
After (Wave II)	21.17	21.03	-0.14
Change (Δ)	-2.16	$+0.59$	

- The DiD estimate:

  $\hat{\tau}_{DiD} = \Delta\bar{y}_{NJ} - \Delta\bar{y}_{PA} = (0.59) - (-2.16) = +2.75$ FTE workers.

- Despite the wage increase, employment in NJ **did not fall** — if anything, it rose slightly relative to PA.
- This famous result challenged the simple competitive model of the labor market and helped launch the modern "credibility revolution" in empirical economics.

DiD and Panel Data

- DiD is closely related to the panel methods of Part I.
- With two periods, the interaction regression is exactly a first-differenced / fixed-effects style design: group fixed effects (T_i) absorb time-invariant group differences, and the time dummy (Post_t) absorbs common trends.
- With **panel data** (same units over time) we can include **unit fixed effects** and **time fixed effects**:

$$y_{it} = a_i + \lambda_t + \tau D_{it} + u_{it},$$

  where $D_{it} = 1$ if unit i is treated in period t. This is the **two-way fixed effects (TWFE)** estimator.
- $\hat{\tau}$ generalizes the 2×2 DiD to many groups and many periods.

Extensions and Cautions

- **Event-study / pre-trends**: estimate effects period-by-period around the treatment date to check that pre-treatment effects are ≈ 0 (supporting parallel trends) and to trace dynamics.
- **Covariates**: add controls so that parallel trends needs to hold only conditionally.
- **Inference**: cluster standard errors at the group level (Bertrand, Duflo, Mullainathan, 2004, warn about serial correlation in DiD).
- **Staggered adoption**: when units are treated at *different* times, the simple TWFE estimator can be biased (recent literature: Goodman-Bacon; Callaway & Sant'Anna; de Chaisemartin & D'Haultfœuille). Newer "robust" DiD estimators address this.
- Threats: differential trends, anticipation effects, spillovers between groups, and compositional changes.

Course Wrap-Up

- **Panel data** let us control for unobserved, time-invariant heterogeneity via first differencing or fixed effects.
- **Causal inference** reframes empirical questions in terms of potential outcomes, treatment effects, and the central problem of constructing a credible counterfactual.
- **Difference-in-differences** is a transparent, widely used quasi-experimental design that uses a control group's trend as the counterfactual, under the parallel trends assumption.
- Together these tools move us from *correlation* toward credible *causal* statements — the goal of modern applied econometrics.

This concludes Econometrics II. Thank you!

References / Further Reading

- Wooldridge, J. *Introductory Econometrics: A Modern Approach* — Ch. 13 (pooled cross sections & panel data, DiD) and Ch. 14 (fixed and random effects).
- Angrist, J. & Pischke, J.-S. (2009). *Mostly Harmless Econometrics*.
- Card, D. & Krueger, A. (1994). "Minimum Wages and Employment." *American Economic Review*.
- Cunningham, S. (2021). *Causal Inference: The Mixtape*.